

## SAFETY DATA SHEET

Creation Date 27-January-2010

Revision Date 02-May-2025

Revision Number 12

### 1. Identification

**Product Name** Dichloromethane

**Cat No. :** AC354800000; AC354800010; AC354800025; AC354800100

**CAS-No** 75-09-2

**Synonyms** Dichloromethane; DCM

**Recommended Use** Laboratory chemicals.

**Uses advised against** .

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

##### **Manufacturer**

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                             |             |
|-------------------------------------------------------------|-------------|
| <b>Skin Corrosion/Irritation</b>                            | Category 2  |
| <b>Serious Eye Damage/Eye Irritation</b>                    | Category 2  |
| <b>Carcinogenicity</b>                                      | Category 1B |
| <b>Specific target organ toxicity (single exposure)</b>     | Category 3  |
| Target Organs - Central nervous system (CNS).               |             |
| <b>Specific target organ toxicity - (repeated exposure)</b> | Category 2  |
| Target Organs - Liver, Kidney, Blood.                       |             |

#### Label Elements

##### **Signal Word**

Danger

**Hazard Statements**

Causes skin irritation  
Causes serious eye irritation  
May cause drowsiness and dizziness  
May cause cancer  
May cause damage to organs through prolonged or repeated exposure

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Wear protective gloves/protective clothing/eye protection/face protection  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Use only outdoors or in a well-ventilated area

**Response**

IF exposed or concerned: Get medical advice/attention  
IF ON SKIN: Wash with plenty of soap and water  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
If skin irritation occurs: Get medical advice/attention  
Take off contaminated clothing and wash it before reuse

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Contains a known or suspected endocrine disruptor

### 3. Composition/Information on Ingredients

| Component       | CAS-No  | Weight % |
|-----------------|---------|----------|
| Dichloromethane | 75-09-2 | >99.5    |

### 4. First-aid measures

**General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ingestion</b>                       | Clean mouth with water and drink afterwards plenty of water.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Most important symptoms/effects</b> | Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Causes central nervous system depression: Continued or high exposures by inhalation will cause anaesthetic effects. This may result in a loss of consciousness and could prove fatal: Causes formation of carbon monoxide in the blood. Carbon monoxide may cause adverse effects on the cardiovascular system and the central nervous system |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

## 5. Fire-fighting measures

|                                         |                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                              |
| <b>Flash Point</b>                      | No information available                                                              |
| <b>Method -</b>                         | No information available                                                              |
| <b>Autoignition Temperature</b>         | 556 °C / 1032.8 °F                                                                    |
| <b>Explosion Limits</b>                 |                                                                                       |
| <b>Upper</b>                            | 23 vol %                                                                              |
| <b>Lower</b>                            | 13 vol %                                                                              |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                              |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                              |

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Phosgene. Hydrogen chloride gas.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
2

**Flammability**  
1

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                                                                                    |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Ensure adequate ventilation. Avoid breathing vapors or mists. Wear respiratory protection.                          |
| <b>Environmental Precautions</b>            | Should not be released into the environment.                                                                                                                       |
| <b>Methods for Containment and Clean Up</b> | Prevent further leakage or spillage if safe to do so. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Ventilate the area. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Vapors are heavier than air and may spread along floors. Handle product only in closed system or provide appropriate exhaust ventilation. Reacts with aluminum and its alloys. |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Do not store in aluminum containers. Incompatible Materials. Strong oxidizing agents. Strong acids. Amines.                                                                                                                            |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component       | Alberta                                   | British Columbia | Ontario TWAEV | Quebec                                    | ACGIH TLV   | OSHA PEL                                                                                                          | NIOSH          |
|-----------------|-------------------------------------------|------------------|---------------|-------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------|----------------|
| Dichloromethane | TWA: 50 ppm<br>TWA: 174 mg/m <sup>3</sup> | TWA: 25 ppm      | TWA: 50 ppm   | TWA: 50 ppm<br>TWA: 174 mg/m <sup>3</sup> | TWA: 50 ppm | (Vacated) TWA: 500 ppm<br>(Vacated) STEL: 2000 ppm<br>(Vacated) Ceiling: 1000 ppm<br>TWA: 25 ppm<br>STEL: 125 ppm | IDLH: 2300 ppm |

#### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

#### Hand Protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments                                         |
|----------------|-------------------|-----------------|--------------------------------------------------------|
| Viton (R)      | < 120 minutes     | 0.7 mm          | As tested under EN374-3                                |
| Nitrile rubber | < 4 minutes       | 0.38 mm         | Determination of Resistance to Permeation by Chemicals |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

In case of inadequate ventilation wear respiratory protection. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** low boiling organic solvent Type AX Brown conforming to EN371

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

No information available.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

Physical State

Liquid

|                                        |                                   |
|----------------------------------------|-----------------------------------|
| Appearance                             | Colorless                         |
| Odor                                   | sweet                             |
| Odor Threshold                         | No information available          |
| pH                                     | Not applicable Insoluble in water |
| Melting Point/Range                    | -97 °C / -142.6 °F                |
| Boiling Point/Range                    | 39 °C / 102.2 °F                  |
| Flash Point                            | No information available          |
| Evaporation Rate                       | No information available          |
| Flammability (solid,gas)               | Not applicable                    |
| Flammability or explosive limits       |                                   |
| Upper                                  | 23 vol %                          |
| Lower                                  | 13 vol %                          |
| Vapor Pressure                         | 350 mbar @ 20°C                   |
| Vapor Density                          | 2.93                              |
| Specific Gravity                       | 1.33                              |
| Solubility                             | No information available          |
| Partition coefficient; n-octanol/water | No data available                 |
| Autoignition Temperature               | 556 °C / 1032.8 °F                |
| Decomposition Temperature              | No information available          |
| Viscosity                              | 0.42 mPas @ 25°C                  |
| Molecular Formula                      | C H <sub>2</sub> Cl <sub>2</sub>  |
| Molecular Weight                       | 84.93                             |

## 10. Stability and reactivity

|                                  |                                                                                          |
|----------------------------------|------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                               |
| Stability                        | Stable under normal conditions. Decomposes on exposure to light.                         |
| Conditions to Avoid              | Excess heat. Protect from direct sunlight.                                               |
| Incompatible Materials           | Strong oxidizing agents, Strong acids, Amines                                            |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Phosgene, Hydrogen chloride gas |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                 |
| Hazardous Reactions              | Forms a detonable mixture with nitric acid.                                              |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

| Component       | LD50 Oral            | LD50 Dermal          | LC50 Inhalation                                            |
|-----------------|----------------------|----------------------|------------------------------------------------------------|
| Dichloromethane | > 2000 mg/kg ( Rat ) | > 2000 mg/kg ( Rat ) | 53 mg/L ( Rat ) 6 h<br>76000 mg/m <sup>3</sup> ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Irritating to eyes and skin                                                              |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component       | CAS-No  | IARC     | NTP                    | ACGIH | OSHA | Mexico |
|-----------------|---------|----------|------------------------|-------|------|--------|
| Dichloromethane | 75-09-2 | Group 2A | Reasonably Anticipated | A3    | X    | A3     |

IARC (International Agency for Research on Cancer)

NTP: (National Toxicity Program)

ACGIH: (American Conference of Governmental Industrial Hygienists)

Mexico - Occupational Exposure Limits - Carcinogens

IARC (International Agency for Research on Cancer)

Group 1 - Carcinogenic to Humans

Group 2A - Probably Carcinogenic to Humans

Group 2B - Possibly Carcinogenic to Humans

NTP: (National Toxicity Program)

Known - Known Carcinogen

Reasonably Anticipated - Reasonably Anticipated to be a Human Carcinogen

A1 - Known Human Carcinogen

A2 - Suspected Human Carcinogen

A3 - Animal Carcinogen

ACGIH: (American Conference of Governmental Industrial Hygienists)

Mexico - Occupational Exposure Limits - Carcinogens

A1 - Confirmed Human Carcinogen

A2 - Suspected Human Carcinogen

A3 - Confirmed Animal Carcinogen

A4 - Not Classifiable as a Human Carcinogen

A5 - Not Suspected as a Human Carcinogen

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Central nervous system (CNS)

**STOT - repeated exposure** Liver Kidney Blood

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Causes central nervous system depression: Continued or high exposures by inhalation will cause anaesthetic effects. This may result in a loss of consciousness and could prove fatal: Causes formation of carbon monoxide in the blood. Carbon monoxide may cause adverse effects on the cardiovascular system and the central nervous system

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** Tumorigenic effects have been reported in experimental animals.

## 12. Ecological information

### Ecotoxicity

| Component       | Freshwater Algae   | Freshwater Fish                           | Microtox                                    | Water Flea         |
|-----------------|--------------------|-------------------------------------------|---------------------------------------------|--------------------|
| Dichloromethane | EC50:>660 mg/L/96h | Pimephales promelas:<br>LC50:193 mg/L/96h | EC50: 1 mg/L/24 h<br>EC50: 2.88 mg/L/15 min | EC50: 140 mg/L/48h |

**Persistence and Degradability** Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its volatility.

| Component       | log Pow |
|-----------------|---------|
| Dichloromethane | 1.25    |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component                 | RCRA - U Series Wastes | RCRA - P Series Wastes |
|---------------------------|------------------------|------------------------|
| Dichloromethane - 75-09-2 | U080                   | -                      |

## 14. Transport information

### DOT

|                      |                 |
|----------------------|-----------------|
| UN-No                | UN1593          |
| Proper Shipping Name | DICHLOROMETHANE |
| Hazard Class         | 6.1             |
| Packing Group        | III             |

### TDG

|                      |                 |
|----------------------|-----------------|
| UN-No                | UN1593          |
| Proper Shipping Name | DICHLOROMETHANE |
| Hazard Class         | 6.1             |
| Packing Group        | III             |

### IATA

|                      |                 |
|----------------------|-----------------|
| UN-No                | UN1593          |
| Proper Shipping Name | Dichloromethane |
| Hazard Class         | 6.1             |
| Packing Group        | III             |

### IMDG/IMO

|                      |                 |
|----------------------|-----------------|
| UN-No                | UN1593          |
| Proper Shipping Name | Dichloromethane |
| Hazard Class         | 6.1             |
| Packing Group        | III             |

## 15. Regulatory information

### International Inventories

| Component       | CAS-No  | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-----------------|---------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Dichloromethane | 75-09-2 | X   | -    | X    | ACTIVE                                        | 200-838-9 | -      | -   |

| Component       | CAS-No  | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------------|---------|-------|----------|------|------|------|------|-------|-------|
| Dichloromethane | 75-09-2 | X     | KE-23893 | X    | X    | X    | X    | X     | X     |

### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component       | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|-----------------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| Dichloromethane | Part 1, Group A Substance Part 4 Substance           | Schedule I                                                                 |                                           |

### Other International Regulations

## Authorisation/Restrictions according to EU REACH

| Component       | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                              | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Dichloromethane | -                                                                   | Use restricted. See entry 59.<br>(see link for restriction details)<br>Use restricted. See entry 75.<br>(see link for restriction details) | -                                                                                                     |

Restricted to industrial use and to approved professionals.

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component       | CAS-No  | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------|---------|----------|------------------------------|---------------------------|--------------------------------------------|
| Dichloromethane | 75-09-2 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component       | CAS-No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Dichloromethane | 75-09-2 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y45                      |

## 16. Other information

## Prepared By

Regulatory Affairs  
Thermo Fisher Scientific  
Email: EMSDS.RA@thermofisher.com

## Creation Date

27-January-2010

## Revision Date

02-May-2025

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02-May-2025

## Revision Summary

This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**